Catching the Spy: Eavesdropper Detection Analysis in Quantum Key Distribution Networks Using Qiskit

Shubham Patel
Department of Computer Science & Engineering
Dr. Bhimrao Ambedkar University, Agra, Uttar Pradesh, India
https://orcid.org/0009-0005-6076-8382
Email: shubhamkumarpatel45@gmail.com

***Abstract*—This paper describes a software-defined simulation of the BB84 quantum key distribution (QKD) protocol implemented with the IBM Qiskit AerSimulator. The pipeline models an authorized sender (Advik), an authorized receiver (Bhaskar), and an unauthorized eavesdropper (Ekta). Advik encodes random bits in the rectilinear (+) or diagonal (×) conjugate basis, Ekta performs a measure-and-resend attack with an independently chosen random basis, and Bhaskar measures the received qubits in a separately chosen random basis. After basis reconciliation over a public classical channel, the quantum bit error rate (QBER) of the sifted key is computed. Monte Carlo runs for 10 to 1000 transmitted qubits give a QBER of 0.00% without interception and a mean of about 25% with interception, in agreement with the theoretical value of 25.00%. The spread of the measured QBER decreases with key length, with a standard deviation of 20.7 percentage points at 10 qubits and 2.0 at 1000 qubits. A fixed rule aborts the session when the QBER exceeds 15%. This rule rejected 73.9% of attacked sessions at 20 qubits and all attacked sessions at 500 and 1000 qubits. A twenty-qubit run with a QBER of 0.00% under interception is explained by a computed probability of about 6.9% for that outcome, which illustrates the weakness of fixed thresholds on short keys.**

***Index Terms*—quantum key distribution, BB84, prepare-and-measure protocol, quantum bit error rate, measure-and-resend attack, Qiskit, eavesdropper detection.**

I. Introduction

Public-key cryptosystems such as RSA [1] and elliptic-curve cryptography (ECC) rely on the computational difficulty of integer factorization and the discrete logarithm problem. Shor showed that both problems can be solved in polynomial time on a sufficiently large quantum computer [2]. Encrypted traffic that is recorded today could therefore be decrypted once such a machine exists. This exposure motivates key-establishment methods whose security rests on physical law and not on computational assumptions.

The BB84 protocol, proposed by Bennett and Brassard in 1984, is a prepare-and-measure protocol for this purpose [3]. A sender encodes each key bit in the polarization of a single photon, using one of two conjugate bases: the rectilinear basis (+), with horizontal and vertical polarization, or the diagonal basis (×), with polarization at 45° and 135°. States from different bases are non-orthogonal, so a measurement in the wrong basis gives a random outcome and cannot reveal the encoded bit with certainty.

Two principles prevent an eavesdropper from observing the photons without detection. The no-cloning theorem states that an unknown quantum state cannot be copied exactly [4]. An eavesdropper therefore cannot keep a copy of a photon and forward the original unchanged. The uncertainty principle [5] limits the simultaneous knowledge of conjugate observables, so a measurement in one basis disturbs the state with respect to the other. A measurement by a third party collapses the wavefunction and introduces errors that the legitimate parties can detect statistically. The security of BB84 against general attacks was proved in an information-theoretic sense by Mayers [6] and by Shor and Preskill [7].

This work implements the protocol in Qiskit [8] and examines how a measure-and-resend attacker affects the error rate of the sifted key. The three parties are named Advik (authorized sender), Bhaskar (authorized receiver), and Ekta (unauthorized eavesdropper). The contributions are a four-stage simulation pipeline, a derivation of the expected error rate under full and partial interception, a statistical analysis of the effect of key length on detection, an analysis of a finite-length case in which the measured QBER is 0.00% despite interception, and an automated abort rule based on a fixed QBER limit. Section II summarizes the theoretical background, Section III reviews related work, Section IV describes the simulator, Section V reports the results, Section VI describes the workflow figure, Section VII discusses the findings, Section VIII lists limitations, and Section IX concludes.

II. Theoretical Background

A. Quantum States and Conjugate Bases

A photon polarization state is modeled as a two-level quantum system [9]. The rectilinear basis consists of the orthogonal states $|0\rangle$ and $|1\rangle$. The diagonal basis consists of $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. The two bases are conjugate: the overlap between any state of one basis and any state of the other has magnitude $1/\sqrt{2}$, so a measurement in the wrong basis yields each outcome with probability 1/2. The encoding used by the protocol is listed in Table I.

TABLE I. BB84 Encoding of Bit Values

Bit	Basis	State	Polarization
0	+	$\|0\rangle$	Horizontal
1	+	$\|1\rangle$	Vertical
0	×	$\|+\rangle$	45°
1	×	$\|-\rangle$	135°

B. Measurement and Disturbance

If the receiver measures in the basis used for encoding, the original bit is recovered with certainty in a noiseless system. If the bases differ, the outcome is uniformly random and independent of the encoded bit. Because the measurement collapses the state into an eigenstate of the measured basis, the

post-measurement state carries no record of the original basis. By the no-cloning theorem [4], an interceptor cannot avoid this disturbance by copying the state before measuring it.

C. Measure-and-Resend Attack

In a measure-and-resend (intercept-resend) attack, the eavesdropper measures every photon in a basis chosen at random and sends a new photon prepared in the state she observed. Consider a position where Advik and Bhaskar used the same basis, so that the position is kept after sifting. The conditional error probabilities are given in Table II. When Ekta chooses the correct basis, which occurs with probability 1/2, the resent photon is identical to the original and no error occurs. When she chooses the wrong basis, also with probability 1/2, the resent photon is in a state of the conjugate basis, and Bhaskar obtains the wrong bit with probability 1/2.

TABLE II. ERROR PROBABILITY ON A SIFTED POSITION

Case	Prob.	Bob error	Contribution
Ekta basis matches	1/2	0	0
Ekta basis differs	1/2	1/2	1/4
Total	1	—	1/4

The expected QBER on the sifted key is therefore 25%. Equivalently, on a sifted position Bhaskar obtains the value that Advik sent with probability 3/4, which is the sum of 1/2 × 1 and 1/2 × 1/2. In the same attack, Ekta's own estimate of each sifted bit is correct in 75% of positions: with certainty when her basis matches, and with probability 1/2 when it does not. The attack thus gives the eavesdropper partial information about the key and, at the same time, produces a measurable error rate for the legitimate parties.

D. Partial Interception

An eavesdropper may intercept only a fraction f of the qubits and leave the others untouched. The expected QBER then scales linearly, QBER = f/4, because only the intercepted positions contribute errors. Ekta knows with certainty a fraction f/2 of the sifted bits, namely the intercepted positions where her basis matched. Table III and Fig. 1 show the relation between f and the expected QBER.

TABLE III. EXPECTED QBER FOR PARTIAL INTERCEPTION

Fraction f	Expected QBER	Bits known to Ekta
0.2	5.0%	10%
0.4	10.0%	20%
0.6	15.0%	30%
0.8	20.0%	40%
1.0	25.0%	50%

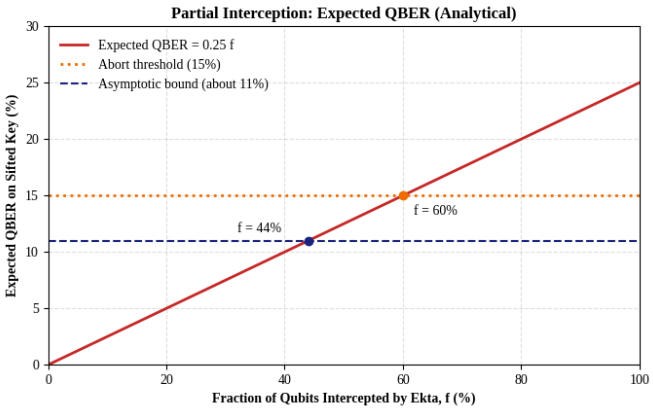


Fig. 1. Expected QBER as a function of the fraction of intercepted qubits (analytical). The dotted line marks the 15% abort threshold and the dashed line marks the asymptotic bound of about 11%.

The linear relation has a consequence for threshold selection. With an abort threshold of 15%, an attack with f below 0.6 produces an expected QBER under the threshold and is not rejected on average, although Ekta gains information on a fraction f/2 of the key. A threshold of 11% corresponds to f of about 0.44. The legitimate parties must therefore remove this partial information by privacy amplification, and the threshold must be chosen with reference to the security analysis.

E. Secret-Key Rate

For BB84 with ideal single-photon sources, the asymptotic secret fraction of the sifted key is given by the Shor–Preskill expression [7]:

$$r = 1 - 2h(Q) \tag{1}$$

where Q is the QBER and h is the binary entropy function, $h(Q) = -Q \log_2 Q - (1 - Q) \log_2 (1 - Q)$. The fraction becomes zero at Q of about 11%, and negative values indicate that no secret key can be extracted. Table IV lists values for representative error rates.

TABLE IV. ASYMPTOTIC SECRET FRACTION FOR BB84

Q	h(Q)	r = 1 − 2h(Q)
0.00	0.0000	1.0000
0.05	0.2864	0.4272
0.11	0.4999	0.0002
0.15	0.6098	−0.2197
0.25	0.8113	−0.6226

At Q = 0.15 the asymptotic fraction is negative. A 15% abort limit therefore functions as a detection and alarm threshold in the simulation, and it does not indicate that a key with a QBER just below 15% would be secure in the asymptotic sense.

F. Post-Processing

A practical protocol follows sifting with error estimation, information reconciliation, and privacy amplification. Error estimation compares a sample of the sifted bits to determine the QBER. Reconciliation corrects the remaining differences between the two keys, for example with the Cascade protocol [10] or the Winnow protocol [11]. Privacy amplification applies a universal hash function to shorten the key and remove the information available to an eavesdropper [12]. The present simulation implements sifting and error estimation only.

G. Threat Model

The analysis uses the following threat model. Ekta has full access to the quantum channel and may measure, replace, or block any photon. She can read all messages on the public classical channel but cannot modify them, because this channel is assumed to be authenticated. She has no access to the devices or the random number generators of Advik and Bhaskar. Her computational power is unrestricted, which is the setting of information-theoretic security. The simulated attack is the full measure-and-resend attack, with the partial attack of Section II-D as a generalization. Attacks that use quantum memory or collective measurements are outside the scope of the simulation.

III. Related Work

Reviews of quantum cryptography summarize the protocols, experimental implementations, and open problems of the field [13], [14], [15], [16], [17]. Experimental demonstrations include satellite-to-ground QKD [18] and entanglement-based QKD over 1120 km [19]. Security analyses of BB84 have progressed from restricted attacks to arbitrary attacks permitted by quantum mechanics [6], [7]. For practical systems, imperfect devices introduce additional weaknesses. An important example is the photon-number-splitting attack, which exploits multi-photon pulses produced by attenuated laser sources [20]. Security proofs that account for imperfect devices are given in [21].

Several protocols modify BB84. The B92 protocol uses only two non-orthogonal states [22]. The SARG04 protocol changes the sifting procedure to reduce the effect of photon-number-splitting attacks [23]. The decoy-state method sends pulses of different mean intensity to detect such attacks [24]. The E91 protocol is based on entangled pairs and on a test of Bell inequalities [25]. When the key is short, the statistical fluctuations of the error estimate must be included in the security bound, and finite-key analyses address this requirement [26]. Software simulators are widely used for teaching and prototyping because they allow the protocol logic to be examined without optical hardware. The simulator described in this paper follows this approach and concentrates on the detection of a single, well-defined attack and on the effect of finite key length.

Table V summarizes the differences among these protocols.

Table V. Comparison of Related QKD Protocols

Protocol	Signal states	Distinguishing feature
BB84	Four states, two bases	Random basis choice and sifting
B92	Two non-orthogonal states	Minimal state set
SARG04	Same states as BB84	Modified sifting
E91	Entangled pairs	Bell-inequality test
Decoy state	BB84 with varied intensity	Detects photon-number splitting

IV. Simulation Architecture

A. Overview

The simulator is a four-stage pipeline executed on the Qiskit AerSimulator (Fig. 2). Each transmitted photon is represented by one qubit. The quantum channel carries the qubits, and an unencrypted but authenticated classical channel is used for public discussion. Authentication of the classical channel is assumed, because an attacker able to modify classical messages could impersonate either party.

B. Stage 1: State Preparation

Advik generates a sequence of uniformly random bits and an independent sequence of uniformly random basis choices. A bit value is mapped to a qubit state by the chosen basis. For a bit value of 1, an X gate is applied to the initial state $|0\rangle$. For the diagonal basis, a Hadamard gate is applied afterwards. This procedure produces the four states of Table I. The gate operations are summarized in Table VI.

C. Stage 2: Interception

When the attack is enabled, Ekta intercepts each qubit and measures it in a randomly chosen basis that is not coordinated with Advik. The measurement collapses the state. Ekta then prepares a new qubit according to her result and sends it to Bhaskar. When Ekta selects the wrong basis, the resent qubit no longer carries the original information. When the attack is disabled, this stage is skipped.

D. Stage 3: Measurement

Bhaskar measures each received qubit in a randomly chosen basis, independent of both Advik and Ekta. For the diagonal basis, a Hadamard gate is applied before the computational-basis measurement. The measurement result is read from the simulator backend.

Table VI. Gate Operations in the Simulation

Operation	Gate sequence
Encode bit 0, + basis	none
Encode bit 1, + basis	X
Encode bit 0, × basis	H
Encode bit 1, × basis	X, then H
Measure in + basis	measure
Measure in × basis	H, then measure

E. Stage 4: Classical Sifting

Advik and Bhaskar announce their basis choices over the public classical channel, without announcing bit values. Positions where the bases differ are discarded. The remaining positions form the sifted key, which has an expected length of one half of the number of transmitted qubits. A subset or the full sifted key is compared to compute the QBER:

$$QBER = N_err / N_sift \quad (2)$$

where N_err is the number of mismatched bits and N_sift is the number of compared bits. If the QBER exceeds the acceptance limit of 15%, the session is aborted and the key is discarded.

F. Simulation Parameters

Table VII lists the parameters of the simulation. The model contains no channel loss and no detector noise, and the source is an ideal single-photon source. Each run uses one circuit of n independent qubits executed with a single shot. The stabilizer method is used because the circuits contain only X, Hadamard, and measurement operations.

Table VII. Simulation Parameters

Parameter	Value
Backend	Qiskit AerSimulator

Transmitted qubits n	10 to 1000
Bases	+ and ×, uniform random
Attack model	Full measure-and-resend
Channel noise	None
Photon source	Ideal single photon
Aer method	Stabilizer
Runs per n	1000 (n ≤ 100), 333 (n ≥ 200)
Random seed	2026
Abort threshold	QBER > 15%

G. Evaluation Metrics

Four quantities are used to evaluate each run. The sifting ratio is the number of kept positions divided by n, which has an expected value of 1/2. The sifted-key length m is a binomial random variable with parameters n and 1/2, with mean n/2 and variance n/4. The probability that no position is kept is 2^(−n), which is negligible for the sequence lengths considered. The QBER is the fraction of mismatched bits among the compared positions. The detection outcome is a binary variable that equals 1 when the QBER exceeds the threshold and the session is aborted. For an attacked channel, the probability that this variable equals 0 is the false-acceptance probability, which is analyzed in Section V-D.

H. Experimental Procedure

For each value of n between 10 and 1000, the secure configuration and the attack configuration are executed with the same pipeline. In the secure configuration, Stage 2 is skipped. In the attack configuration, Stage 2 is applied to every qubit. The QBER is computed from the sifted key after Stage 4, and the abort rule is applied. The resulting values are compared with the theoretical value of 25% and with the 15% threshold. Each configuration is repeated 1000 times for n up to 100 and 333 times for larger n, and the QBER of every run is recorded (Section V-B). The analytical probabilities of Section V-D are compared with the observed frequencies.

I. Algorithm Summary

1. For each of n qubits, Advik draws a random bit and a random basis, and prepares the corresponding state.
2. If the attack is enabled, Ekta draws a random basis, measures the qubit, and prepares a new qubit from her result.
3. Bhaskar draws a random basis and measures the received qubit.
4. Advik and Bhaskar publish their bases and keep the positions with equal bases.
5. The QBER is computed from the kept positions.
6. If QBER > 15%, the key is discarded and the session is aborted; otherwise the key is accepted.

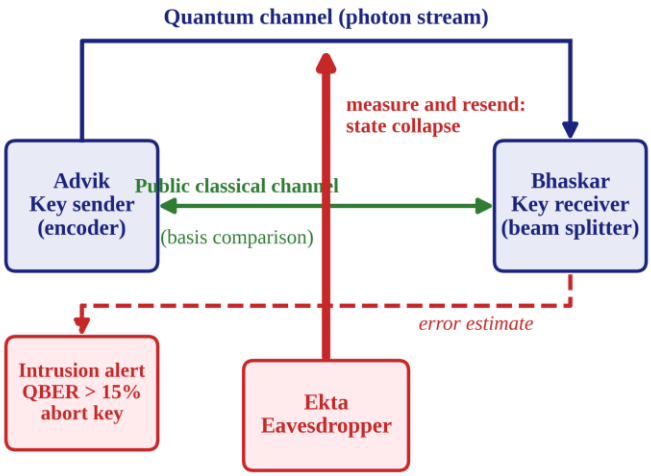


Fig. 2. Workflow of the simulated BB84 network with an eavesdropper. The blue link is the quantum channel, the green link is the public classical channel, and the red elements show the interception and the intrusion alert.

J. Worked Example

Table VIII gives an illustrative example with eight qubits, constructed to explain the stages and not obtained from the simulator. In each row, A, E, and B denote Advik, Ekta, and Bhaskar. When the basis of Ekta equals that of Advik, her result equals the bit sent. When it differs, her result is random, and the qubit that she resends is in her measured basis. Bhaskar's result is correct when his basis equals that of Advik and the resent state is in the same basis, and it is random otherwise. Positions 1, 2, 4, 5, 6, and 8 are kept after sifting. Position 2 contains an error, so the sifted key of six bits has a QBER of 1/6, or 16.67%. This value exceeds the 15% threshold, and the session is aborted. Positions 6 and 8 show cases in which Ekta used the wrong basis and no error appeared by chance.

TABLE VIII. ILLUSTRATIVE EIGHT-QUBIT EXAMPLE

#	A bit	A bas.	E bas.	E res.	B bas.	B res.	Kept	Err.
1	0	+	+	0	+	0	yes	no
2	1	×	+	0	×	0	yes	yes
3	1	+	×	0	×	1	no	—
4	0	×	×	0	×	0	yes	no
5	1	+	+	1	+	1	yes	no
6	0	+	×	1	+	0	yes	no
7	0	×	×	0	+	0	no	—
8	1	×	+	1	×	1	yes	no

K. Software Environment and Reproducibility

The Monte Carlo experiments used Python 3.12.3, Qiskit 2.6.1, and Qiskit Aer 0.18.0 with the stabilizer method, which simulates circuits of Clifford operations efficiently for up to 1000 qubits. A fixed pseudorandom seed (2026) initializes the generator for bits and bases, and each simulator execution receives its own seed from this generator. The script bb84_sim.py implements the pipeline and is provided as supplementary material. Runs in which no position was kept after sifting (m = 0) were excluded, which affected two attacked runs and one secure run at n = 10 and no run at larger n. The single-run results of Section V-A were obtained separately from the Monte Carlo experiments.

V. Results

A. Secure Channel and Attack Scenarios

Table IX lists the simulated outcomes. In the secure configuration, no eavesdropper is present, and Advik and Bhaskar obtain identical sifted keys. The QBER is 0.00% in every run, as shown by the baseline in Fig. 3. This result is expected for an ideal noiseless simulator.

TABLE IX. SIMULATED QBER AND SESSION OUTCOME

Configuration	QBER	Session outcome
Secure channel (no Ekta)	0.00%	Key accepted
Measure-and-resend, theoretical	25.00%	Key aborted
Measure-and-resend, short run	36.36%	Key aborted

In single runs under the measure-and-resend attack, the simulated error rates in Fig. 3 lie close to the theoretical value of 25% for 10 to 50 transmitted qubits, with values between about 22% and 26%. The agreement supports the derivation in Section II-C.

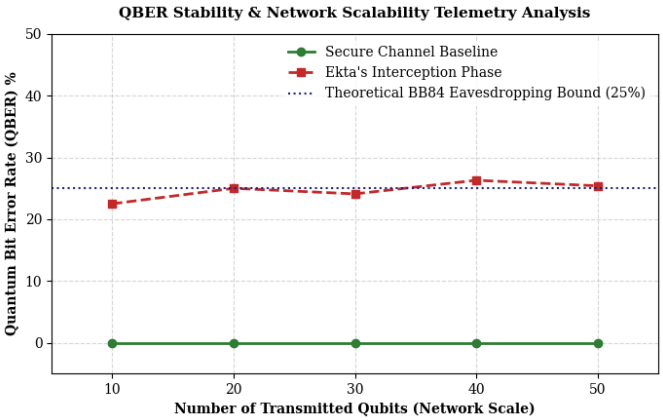


Fig. 3. QBER of the secure channel and of the intercepted channel for 10 to 50 transmitted qubits, with one run per value of n. The dotted line marks the theoretical 25% bound for a measure-and-resend attack.

B. Monte Carlo Evaluation

To characterize the spread of the QBER, each configuration was repeated many times for n from 10 to 1000. Table X summarizes the results for the attacked channel, and Fig. 4 shows the mean and the standard deviation. In the secure configuration, the QBER was 0.00% in every run for every n.

TABLE X. MONTE CARLO RESULTS UNDER FULL INTERCEPTION

n	Runs	Mean QBER (%)	S.D. (%)	95% range (%)	Aborted (%)
10	998	24.58	20.71	0.0–66.7	68.7
20	1000	25.38	14.59	0.0–55.6	73.9
30	1000	24.76	11.46	5.9–50.0	79.2
40	1000	25.02	9.68	7.1–45.5	84.0
50	1000	24.76	8.43	9.1–42.9	88.5
100	1000	24.93	6.13	13.9–37.5	95.7
200	333	25.37	4.14	17.5–33.2	98.8
500	333	24.88	2.89	19.8–31.0	100
1000	333	24.86	1.98	20.9–28.6	100

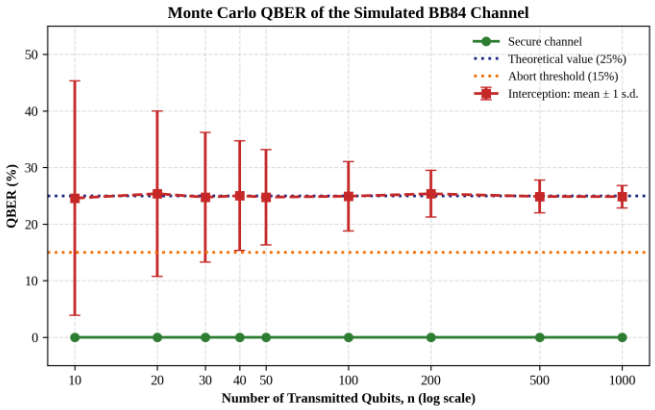


Fig. 4. Mean QBER and one standard deviation from Monte Carlo runs (333 to 1000 runs per value of n) for the secure channel and the channel with full interception.

The mean QBER under attack lies between 24.6% and 25.4% for every n, in agreement with the theoretical value of 25%. The standard deviation decreases with n and follows the binomial prediction $\sqrt{(0.25 \times 0.75/(n/2))}$, which gives 6.1% for n = 100 and 1.9% for n = 1000, against observed values of 6.13% and 1.98%. The abort rule rejected 68.7% of attacked sessions at n = 10, 73.9% at n = 20, 88.5% at n = 50, and 95.7% at n = 100. It rejected all attacked sessions at n = 500 and n = 1000. Short sequences therefore remained exposed to the false-acceptance problem analyzed in Section V-D.

C. Statistical Variation with Key Length

The number of compared bits determines the spread of the measured QBER. For m compared bits and a true error probability p = 0.25, the number of errors follows a binomial distribution, and the standard deviation of the measured QBER is $\sqrt{(p(1 - p)/m)}$. Because only about half of the transmitted qubits remain after sifting, a run of 10 qubits yields roughly 5 compared bits and a standard deviation near 19%. One simulation run produced a QBER of 36.36%, which exceeds the theoretical mean. This value corresponds, for example, to 4 mismatches among 11 sifted bits, and it is consistent with the large variance expected for such short keys. Table XI gives the standard deviation and the probability that an attacked session passes the 15% test for several values of m.

TABLE XI. EFFECT OF SIFTED-KEY LENGTH UNDER ATTACK

m (bits)	Std. dev. of QBER	P(QBER ≤ 15%) under attack
5	19.4%	23.7%
11	13.1%	19.7%
25	8.7%	9.6%
50	6.1%	4.5%
100	4.3%	1.1%

The last column shows that a short key can pass the test by chance, with a probability of about one quarter for m = 5. The probability falls below 5% at m = 50 and to about 1% at m = 100. The tail probabilities of a sample proportion can also be bounded with Hoeffding's inequality [27], which shows that the estimation error decreases with the square root of the sample size. A related quantity is the probability that at least one error appears among m compared bits, which equals 1 − (3/4)^m. This value is 0.944 for m = 10 and exceeds 0.999 for m = 25.

Hoeffding's inequality also gives a sample size for a required accuracy. The probability that the measured QBER deviates from the true value by at least ε is bounded by $2\exp(-2m\varepsilon^2)$. Setting this bound equal to a confidence parameter δ gives $m \geq \ln(2/\delta)/(2\varepsilon^2)$. Table XII lists the required number of compared bits for several choices. Since only half of the transmitted qubits are kept, the number of transmitted qubits is about twice the values listed. The sequences of 10 to 50 qubits used in the simulation are far below these requirements.

TABLE XII. COMPARED BITS REQUIRED FOR A QBER ESTIMATE

ε	δ	Required m (bits)
0.10	0.05	185
0.10	0.01	265
0.05	0.05	738
0.05	0.01	1060

D. Finite-Length Boundary Case at 0.00% QBER

In the quantitative evaluation of the intercept-resend configuration, one run with twenty transmitted qubits produced a QBER of exactly 0.00% although every photon had been intercepted. This outcome does not contradict the physics of the protocol. An unauthorized measurement on non-orthogonal states perturbs the system and collapses the wavefunction, but the perturbation produces an error only with a certain probability. On a sifted position, Bhaskar obtains the value sent by Advik with probability 3/4, as shown in Section II-C. A finite sequence can therefore contain no mismatch by chance.

The probability of this event can be computed. A transmitted qubit contributes an error to the sifted key with probability 1/2 × 1/4 = 1/8, because the bases of Advik and Bhaskar must match (probability 1/2) and an error must then occur (probability 1/4). The qubits are independent, so the probability that no error appears among n qubits is (7/8)^n. For n = 20 this probability is 0.069, or about 6.9%. The event is uncommon but not negligible, and it is more likely for shorter sequences. Table XIII gives the probability of a 0.00% QBER under attack and the probability that the attacked session is accepted by the 15% rule, and Fig. 5 shows both quantities as a function of n.

The probability that the attacked session is accepted by the threshold rule follows from the distribution of the sifted-key length m and of the number of errors k. It is given by

$$P_{acc}(n) = \sum_m B(m; n, 1/2) \cdot F(\lfloor 0.15m \rfloor; m, 1/4) \quad (3)$$

where B is the binomial probability mass function, F is the binomial cumulative distribution function, and the sum runs over m from 1 to n. Sessions with m = 0 yield no key and are excluded. The values in Table XIII were computed from this expression. The simulated frequencies agree with the model within the sampling error. For n = 20, the QBER was 0.00% in 7.6% of 1000 attacked runs, against a computed probability of 6.92%.

TABLE XIII. UNDETECTED-ATTACK PROBABILITY: MODEL AND SIMULATION

n	QBER = 0%, model	QBER = 0%, sim.	Accepted, model	Accepted, sim.
10	26.3%	27.3%	31.3%	31.3%
20	6.92%	7.6%	25.5%	26.1%
30	1.82%	2.2%	19.5%	20.8%
40	0.48%	0.4%	16.3%	16.0%
50	0.13%	0.1%	12.9%	11.5%
100	0.0002%	0.0%	4.8%	4.3%
200	≈ 0	0.0%	0.84%	1.2%

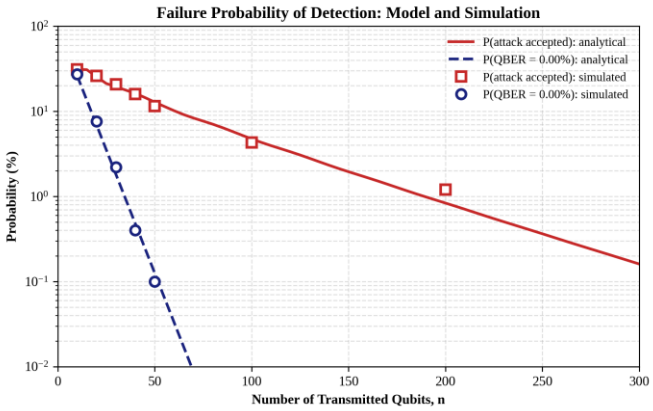


Fig. 5. Probability that a full measure-and-resend attack yields a QBER of 0.00% (dashed) or a QBER not above the 15% threshold (solid): analytical curves and simulated frequencies (markers).

This result demonstrates a vulnerability of finite-length protocols. A QBER of zero in a short run does not prove that the channel was not intercepted, and a fixed threshold applied to a small sample gives a non-zero probability of accepting a compromised key. The probability that the attack remains undetected decreases with the length of the sequence, as shown in Fig. 5, but it does not reach zero for any finite n. Information-theoretic security therefore requires that the number of transmitted qubits be large enough for the statistical uncertainty to be included in the security bound [26], and that the final key be shortened by privacy amplification [12]. For the short sequences used in this study, the 0.00% outcome should be treated as a statistical fluctuation.

E. Abort Decision

Fig. 6 compares the error rate attributed to each party in the short run against the 15% abortion threshold. The values for the attacked configurations are above the threshold, and the decision rule aborts the session in each such case. The sender's own baseline lies below the threshold.

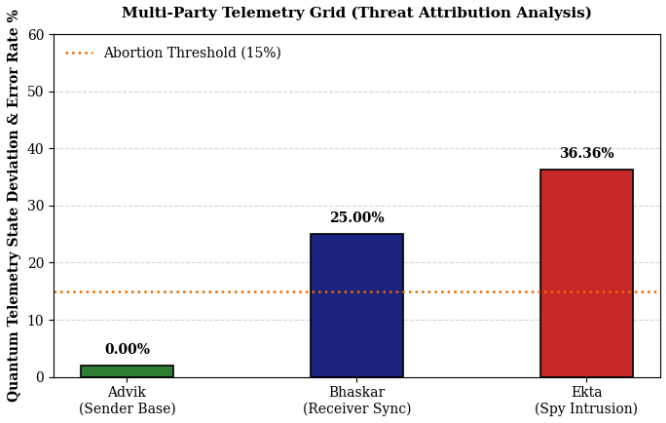


Fig. 6. Quantum state deviation and error rate by party, compared with the 15% abortion threshold (dotted line).

The 15% limit is an operating parameter of this simulation. The asymptotic error bound under which BB84 can still produce a secret key is approximately 11% for ideal single-photon sources and unbounded key length [7], as shown in Table IV. A deployed system would select its threshold according to the finite key length and the security parameter required.

F. Key Yield and Information Gained by Ekta

The yield of the protocol decreases at each stage. Sifting removes half of the positions on average. If half of the sifted bits are disclosed for error estimation, one quarter of the transmitted qubits remains as raw key before reconciliation and privacy amplification. Table XIV gives the expected values for the sequence lengths of the simulation. When a session is aborted, the yield is zero.

TABLE XIV. EXPECTED KEY YIELD PER SEQUENCE LENGTH

n	Expected sifted bits	Raw key after 50% disclosure
10	5	2.5
20	10	5
30	15	7.5
40	20	10
50	25	12.5

The yield for these lengths is small, which agrees with the observation in Section V-C that the error estimate is unreliable. The public announcement of the bases also benefits the eavesdropper. After the announcement, Ekta knows which of her measurement results were obtained in the basis used by Advik, and these results equal the sifted bits exactly. Under full interception, she therefore knows half of the sifted key with certainty and has no information on the other half. This information is removed in a complete protocol by privacy amplification [12], which requires the QBER estimate to be reliable.

VI. VISUAL REPRESENTATION OF THE WORKFLOW

Fig. 2 shows the workflow of the simulated network. Advik is placed on the left as the key sender, with the encoder blocks that generate photon states. Bhaskar is placed on the right as the key receiver, with the beam-splitter blocks used for basis measurement. The upper link is the quantum channel, which carries the photon stream. The central link is the classical channel, over which the parties verify which photons were measured in matching bases. Ekta is placed at the bottom center. A thick red vertical arrow shows her intervention in the quantum channel, which alters the photon states and triggers the detection and alert loop returning to Advik.

VII. DISCUSSION

The simulation confirms the main qualitative property of BB84: interception produces an error rate that the legitimate parties can measure. The mean error rate agrees with the value of 25% derived from the conditional probabilities of Table II. The individual runs, however, show large deviations, For n = 10, the 95% range of the QBER in the Monte Carlo runs extends from 0.00% to 66.7%, and single runs of 0.00% and 36.36% were also observed. These deviations are consistent with the binomial model and are caused by the small number of sifted bits.

Three points follow for the design of a detection rule. First, a threshold test on a single short run has a false-acceptance probability that is far from negligible, as Table XIII shows. Second, a threshold below the expected QBER of full interception does not detect partial interception reliably, as Table III shows, since the expected QBER of an attack on a fraction f is f/4. Third, a QBER under the threshold does not guarantee secrecy, since the asymptotic secret fraction of Table IV is zero at about 11%. For these reasons, a practical system combines a threshold test with a sufficiently long sample, privacy amplification, and a security bound that includes finite-size effects [26].

The relation between the simulation and physical implementations is limited. The simulation isolates the information-disturbance trade-off, which is the central mechanism of BB84. Physical systems add photon loss, dark counts, detector inefficiency, and imperfect sources, which are treated in the security literature [14], [21].

Several practical guidelines follow from the results. The number of bits used for error estimation should be chosen from a required accuracy and confidence, as in Table XII. The abort threshold should exceed the baseline error of the physical channel by a margin, so that honest sessions are not rejected, and it should remain below the security bound of Table IV. An aborted session should be repeated with new random bits and new bases, and the discarded data should not be reused. The result of the threshold test should not be interpreted as proof of the absence of an eavesdropper, because both a short run and a partial attack can pass the test.

VIII. LIMITATIONS

Several limitations apply. First, the simulator is ideal. It contains no channel loss, detector noise, or imperfect state preparation, so the baseline QBER is exactly zero. In a physical system, a nonzero baseline error exists, and the detection rule must distinguish it from the effect of an attacker. Second, only one attack is modeled. The measure-and-resend attack is a basic individual attack, and more general collective and coherent attacks are covered by the security proofs cited above rather than by this simulation [6], [7]. Third, the sources are ideal single-photon sources, so photon-number-splitting attacks [20] cannot occur. Fourth, the single-run results rely on short sequences of 10 to 50 qubits, and the analysis shows that such sequences give unreliable error estimates. The Monte Carlo experiments extend to 1000 qubits but use a limited number of runs per value of n. Fifth, the classical channel is assumed to be authenticated, and the random number generators are assumed to be ideal. Sixth, the curves in Figs. 1 and 5 are analytical results, and only the markers in Fig. 5 are outputs of the simulator.

IX. FUTURE WORK AND CONCLUSION

Several extensions of the simulator are planned. The first is the addition of a channel noise model, such as depolarizing or Gaussian thermal noise, so that errors from the channel can be separated from errors caused by an eavesdropper. The second is the integration of classical error-correction protocols, namely

Cascade [10] and Winnow [11], followed by privacy amplification [12]. The third is an extension to entanglement-based protocols, starting with E91 [25], and then to multi-party settings. A fourth extension is a set of partial-interception experiments that test the relation QBER = f/4, together with a finite-key security bound. A fifth extension is the implementation of weak-pulse sources and the decoy-state method [24].

This study implemented the BB84 protocol in Qiskit and simulated a measure-and-resend attack by an unauthorized third party. The QBER was 0.00% without interception and close to 25% with interception in Monte Carlo experiments of 333 to 1000 runs per value of n, in agreement with the theoretical value derived in Section II. The 15% rule aborted all attacked sessions at 500 and 1000 qubits, but it accepted about 31% of attacked sessions at 10 qubits and 26% at 20 qubits. The analysis also showed that a short sequence can yield a QBER of 0.00% under full interception, with a probability of about 6.9% for twenty qubits, and that short keys can pass a fixed threshold by chance. The security parameter must therefore be scaled to longer sequences, and finite-size effects must be included in the security bound. The results indicate that a software-defined quantum pipeline can reproduce the detection behavior of BB84 and its finite-length weakness in a controlled testbed. The simulation assumes ideal devices and no channel noise, and results for physical systems will differ.

Appendix: State-Vector Calculation

Let Advik encode the bit 0 in the diagonal basis, which gives the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Suppose that Ekta measures in the rectilinear basis. The outcomes $|0\rangle$ and $|1\rangle$ occur with probability $|\langle 0|+\rangle|^2 = 1/2$ and $|\langle 1|+\rangle|^2 = 1/2$. After the measurement, the state is $|0\rangle$ or $|1\rangle$, and Ekta resends this state. Bhaskar measures in the diagonal basis, which matches the basis of Advik. For the resent state $|0\rangle$, the probability of the outcome $|+\rangle$, which gives the correct bit 0, is $|\langle +|0\rangle|^2 = 1/2$, and the probability of $|-\rangle$, which gives the wrong bit 1, is $|\langle -|0\rangle|^2 = 1/2$. The same values hold for the resent state $|1\rangle$. The error probability for this case is therefore 1/2.

If Ekta measures in the diagonal basis, the state $|+\rangle$ is an eigenstate of her measurement. She obtains the outcome $|+\rangle$ with probability 1, resends $|+\rangle$, and Bhaskar obtains the correct bit with probability 1 in a noiseless system. By the law of total probability, the error probability on a sifted position is $1/2 \times 0 + 1/2 \times 1/2 = 1/4$. The calculation is symmetric for the other three states of Table I. If a fraction f of the qubits is intercepted, the expected error probability is f/4.

Author Contributions

Shubham Patel designed the simulation, wrote the software, performed the experiments and analysis, and wrote the manuscript.

Funding

This research received no external funding.

Conflict of Interest

The author declares no conflict of interest.

Use of Generative AI

Generative AI tools were used to assist with drafting the text and with preparing figures. The author reviewed and edited all content and is responsible for the manuscript.

Data Availability

The simulation script (bb84_sim.py), the analysis script, and the Monte Carlo output are provided as supplementary material.

Acknowledgment

The author thanks the Department of Computer Science & Engineering, Dr. Bhimrao Ambedkar University, Agra, for academic support.

References

[1] R. L. Rivest, A. Shamir, and L. Adleman, "A method for obtaining digital signatures and public-key cryptosystems," Commun. ACM, vol. 21, no. 2, pp. 120–126, 1978.

[2] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," SIAM J. Comput., vol. 26, no. 5, pp. 1484–1509, 1997.

[3] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in Proc. IEEE Int. Conf. Computers, Systems and Signal Processing, Bangalore, India, 1984, pp. 175–179.

[4] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," Nature, vol. 299, pp. 802–803, 1982.

[5] W. Heisenberg, "Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik," Z. Phys., vol. 43, pp. 172–198, 1927.

[6] D. Mayers, "Unconditional security in quantum cryptography," J. ACM, vol. 48, no. 3, pp. 351–406, 2001.

[7] P. W. Shor and J. Preskill, "Simple proof of security of the BB84 quantum key distribution protocol," Phys. Rev. Lett., vol. 85, no. 2, pp. 441–444, 2000.

[8] Qiskit contributors, "Qiskit: An open-source framework for quantum computing," Zenodo, 2023, doi: 10.5281/zenodo.2573505.

[9] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information. Cambridge, U.K.: Cambridge Univ. Press, 2000.

[10] G. Brassard and L. Salvail, "Secret-key reconciliation by public discussion," in Advances in Cryptology—EUROCRYPT '93, 1994, pp. 410–423.

[11] W. T. Buttler, S. K. Lamoreaux, J. R. Torgerson, G. H. Nickel, C. H. Donahue, and C. G. Peterson, "Fast, efficient error reconciliation for quantum cryptography," Phys. Rev. A, vol. 67, no. 5, p. 052303, 2003.

[12] C. H. Bennett, G. Brassard, and J.-M. Robert, "Privacy amplification by public discussion," SIAM J. Comput., vol. 17, no. 2, pp. 210–229, 1988.

[13] N. Gisin, G. Ribordy, W. Tittel, and H. Zbinden, "Quantum cryptography," Rev. Mod. Phys., vol. 74, no. 1, pp. 145–195, 2002.

[14] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, "The security of practical quantum key distribution," Rev. Mod. Phys., vol. 81, no. 3, pp. 1301–1350, 2009.

[15] H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," Nature Photon., vol. 8, no. 8, pp. 595–604, 2014.

[16] S. Pirandola et al., "Advances in quantum cryptography," Adv. Opt. Photon., vol. 12, no. 4, pp. 1012–1236, 2020.

[17] F. Xu, X. Ma, Q. Zhang, H.-K. Lo, and J.-W. Pan, "Secure quantum key distribution with realistic devices," Rev. Mod. Phys., vol. 92, no. 2, p. 025002, 2020.

[18] S.-K. Liao et al., "Satellite-to-ground quantum key distribution," Nature, vol. 549, pp. 43–47, 2017.

[19] J. Yin et al., "Entanglement-based secure quantum cryptography over 1,120 kilometres," Nature, vol. 582, pp. 501–505, 2020.

[20] G. Brassard, N. Lütkenhaus, T. Mor, and B. C. Sanders, "Limitations on practical quantum cryptography," Phys. Rev. Lett., vol. 85, no. 6, pp. 1330–1333, 2000.

[21] D. Gottesman, H.-K. Lo, N. Lütkenhaus, and J. Preskill, "Security of quantum key distribution with imperfect devices," Quantum Inf. Comput., vol. 4, no. 5, pp. 325–360, 2004.

[22] C. H. Bennett, "Quantum cryptography using any two nonorthogonal states," Phys. Rev. Lett., vol. 68, no. 21, pp. 3121–3124, 1992.

[23] V. Scarani, A. Acín, G. Ribordy, and N. Gisin, "Quantum cryptography protocols robust against photon number splitting attacks for weak laser pulse implementations," Phys. Rev. Lett., vol. 92, no. 5, p. 057901, 2004.

[24] H.-K. Lo, X. Ma, and K. Chen, "Decoy state quantum key distribution," Phys. Rev. Lett., vol. 94, no. 23, p. 230504, 2005.
[25] A. K. Ekert, "Quantum cryptography based on Bell's theorem," Phys. Rev. Lett., vol. 67, no. 6, pp. 661–663, 1991.
[26] M. Tomamichel, C. C. W. Lim, N. Gisin, and R. Renner, "Tight finite-key analysis for quantum cryptography," Nature Commun., vol. 3, p. 634, 2012.
[27] W. Hoeffding, "Probability inequalities for sums of bounded random variables," J. Amer. Statist. Assoc., vol. 58, no. 301, pp. 13–30, 1963.